PROCEDURE

- The Vernier caliper is checked for zero errors and error if found is to be noted.
- The sphere is kept in between the jaws of the Vernier caliper and the main scale reading (MSR)is noted.
- Vernier scale division which coincides with some main scale division (VSC) is noted. Multiply this VSC by least count (LC) gives Vernier scale reading (VSR).
- Add MSR with VSR. This will be the diameter of the sphere.
- Observations are to be recorded for different positions of the sphere and the average value of the diameter is found. From this value radius of the sphere R is calculated.
- Using the known value of the mass of the sphere M and calculated radius of the sphere R the moment of inertia of the given sphere about its diameter can be calculated using the given formula.

LEAST COUNT (LC)

One main scale division (MSD) = ... cm

Number of Vernier scale divisions = ...

$$\text{Least Count (LC)} = \frac{\text{1 Main Scale Division (MSD)}}{\text{Total Vernier scale divisions}}$$

= ..... cm

OBSERVATIONS

Zero error =

Zero correction (Z . C) = Zero error × LC

Sl.No.	MSR $\times 10^{-2}$m	Vernier coincidence VSC (div)	VSR = (VSC × LC) $\times 10^{-2}$m	TR = (MSR +VSR) $\times 10^{-2}$m	Diameter of the sphere = 2R correct reading = (TR ± Z.C) $\times 10^{-2}$m
1					
2					
3					
4					
5					
6					

Mean diameter 2R = ... m

Radius of the sphere R = ... m

R = ... m

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